EMBAI, midterm summary (BMEVIMIMA22). 2026-10-29, 18:00 to 20:00. 50 points in 6 exercises, signature at 20. TRUE/FALSE: +1 correct, −1 wrong, 0 if blank, total rounded up to 0. Expected value of an answer = $2p-1$: answer only when clearly sure. Write the conventions on the first line (error function, positive class). Calculation required: the result alone earns nothing. Calculator with e^x and ln; $\log_2 x = \ln x / \ln 2$.

1. Recurring TRUE/FALSE items

Lectures 1 and 2

F $O(N^2)$ is less costly than $O(\log_2 N)$ for large N.
T $O(\log_3 N)$ and $O(\log_2 N)$ are the same class (factor $\log_2 3 \approx 1.585$); splitting into thirds still needs fewer weighings.
F The brain beats the computer because its cycle time is shorter. A neuron takes about 1 ms; the advantage lies in parallelism and the 20 W.
F In Edge AI, the computations are done in the cloud (inference on the device).
T Intelligent solutions are needed: the state space is too large for brute force.
F A rational agent always takes the decision that turns out best afterwards (it maximizes the expected utility).
T A rational agent can take a decision that later leads to a loss.
F Rational means choosing the solution with the shortest execution time.
T Quantization: reduce the precision of the weights; pruning: remove weights or neurons.

Cloud or local: neither solves the local energy limit; local improves the reaction time; cloud helps with memory. CPU latency oriented, GPU throughput oriented.

Information and trees

T Four equiprobable directions: 2 bits.
F Two directions: 2 bits (at most 1).
T Two outcomes with no prior information: 1 bit.
F The top card is a diamond: 1 bit ($I(\frac{1}{4}, \frac{3}{4}) = 0.811$).
T Four equiprobable colours need more information than man or woman (2 against at most 1).
F Three equiprobable colours: exactly 3 bits ($\log_2 3 = 1.585$).
F A node reached by only one class: 1 bit (0 bit).
F A node with equal counts of both classes: zero need (1 bit, the maximum).
F The trivial tree generalizes well (it memorizes; the form "does not generalize" is T).
T An attribute with a unique value for each example is useless as a test (maximal gain, but memorization).
F Such an attribute (serial number) is the best test at the root.
T The gain does not depend only on the need before and after the test (also on the weight $(p_k + n_k)/(p+n)$).
F 10 binary attributes all tested on one path: the leaf is certainly homogeneous (label noise).
T Tree construction is greedy. **F**: it is a breadth-first search. **F**: it always finds the smallest tree.
T A tree can represent any logical function of its attributes (sometimes exponential size).

Metrics, ROC, validation

F Accuracy measured on different, large sets of the same size identifies the best method (class proportions).
F 92% accuracy is necessarily better than 80% (example B of lecture 3, useless).
T Sensitivity 1: all positives detected ($FN = 0$); nothing about FP.
F AUC is typically 1 for random guessing (0.5). **F**: AUC of an irrelevant attribute = 1.
T AUC 0.83 against 0.75: the second attribute is better.
T AUC 0.3: worse than random.
F Sensitivity is always lower than specificity (lecture: 0.8 and 0.55).
T Declaring everything positive: TPR = 1, TNR = 0 (ZH 2021 item badly worded: blank possible).
T FPR = 1 − specificity.
F The validation set measures the final performance (that is the test set).
F Against overfitting, the training error is monitored (validation).
T The training error can still decrease during overfitting. **F**: it no longer decreases.
T Less complex tools are less prone to overfitting (and conversely).
F 8 examples, MLP with 5 layers of 10: sensible (hundreds of weights).
F Adding hidden neurons always lowers the test error.

Clustering

T Very different standard deviations: Mahalanobis preferable to Euclidean.
F K-means does not depend on the initial centroids.
T Cost of K-means $O(NKJd)$.
F Bottom-up hierarchical clustering requires the number of clusters in advance.
F Single and complete linkage always give the same merges.
T DBSCAN isolates noise without fixing the number of clusters.
F PCA captures nonlinear relationships.

Neural networks

F The slope of ReLU is always below 1 (or below 0.5): it is 0 or 1.
F The derivative of ReLU is +1 or −1 (0 or +1).
T The slope of the sigmoid is always below 1 (≤ 0.25).
T In backpropagation, the derivatives of the sigmoids are multiplied together.
T BP computes the sensitivity of the error to each weight with the chain rule.
F Moving in the direction of the gradient lowers the error (opposite: $-\nabla E$).
F Moving along $-\nabla E$ first raises the error.
F Moving perpendicular to the gradient certainly raises the error.
T 2000 neurons to 50: on the order of 100,000 weights.
F Fully connected MLP with three layers of 20: at most 60 weights (400 per connection).
T A single perceptron is error-free only if linearly separable.
F Hilbert's 13th problem gives the required number of neurons.
T Without a nonlinear activation, a multilayer network is no more powerful than one layer.
F Fall detection use case: int8 more accurate than float32 (552 against 555 out of 561).

2. One training step (series 1 ex. 2, quiz 06.3)

$s = \sum_k w_k x_k$ with $x_0 = 1$ for the bias; $y = f(s)$; $\varepsilon = d - y$.

$$\frac{\partial E}{\partial w_k} = \frac{\partial E}{\partial \varepsilon}\frac{\partial \varepsilon}{\partial y}\frac{\partial y}{\partial s}\frac{\partial s}{\partial w_k} = c\,\varepsilon \cdot (-1) \cdot f'(s) \cdot x_k$$

$c = 1$ if $E = \frac{1}{2}\varepsilon^2$ (lab), $c = 2$ if $E = \varepsilon^2$ (past paper solutions).

$$w_k \leftarrow w_k - \alpha \frac{\partial E}{\partial w_k} = w_k + \alpha\, c\, \varepsilon\, f'(s)\, x_k$$

Output: $\delta_{\text{out}} = c\,\varepsilon\, f'(s_{\text{out}})$. Hidden j: $\delta_j = f'(s_j) \sum_k \delta_k w_{kj}$ (old weights). Update: $\Delta w = \alpha\, \delta \times$ input carried by this weight (1 for a bias, x_i, or h_j for a hidden to output weight). Quiz 06.2: $\frac{\partial y}{\partial w_k} = f'(s)\, x_k$. Threshold perceptron (output ±1): $w \leftarrow w + \alpha\, \varepsilon\, x$, $\varepsilon \in \{0, \pm 2\}$.

Method.

1. Write E and $\varepsilon = d - y$.
2. Forward pass, neuron by neuron: s, then $f(s)$; 4 decimals.
3. f': sigmoid $y(1-y)$; tanh $1 - y^2$; ReLU 0 or 1; s^2 gives $2s$; linear 1.
4. δ_{out}, then δ_j of the hidden neurons with the old output weights.
5. $w_{\text{new}} = w + \alpha\, \delta \times$ input. Check: $y < d$, input and $f' > 0$: the weight goes up.

Series 1 ex. 2 ($d = 0.9$, $\alpha = 0.1$, new w_{01}). $s_1 = 0.6$, $h_1 = 0.6457$; $s_2 = 0.85$, $h_2 = 0.7006$; $s_{\text{out}} = -0.5063$, $y = 0.3761$; $\varepsilon = 0.5239$; $f'(s_{\text{out}}) = 0.2347$; $f'(s_1) = 0.2288$.

	$E = \varepsilon^2$	$E = \frac{1}{2}\varepsilon^2$
δ_{out}	0.2459	0.1229
$\delta_1 = \delta_{\text{out}}\, w_1\, f'(s_1)$	0.01688	0.00844
new w_{01}	**0.8017**	**0.8008**
new w_1 (variant V5)	0.3159	0.3079

Hidden2 plays no role in w_{01}; $w_{11} = 0$ and $x_1 = 5$ are distractors.

Past papers. ZH 2023, $y = s^2$, weights $(0.5; 0.2; 2.5)$, $x = (1; 0.7; -0.2)$, $d = 0.2$, $\alpha = 0.1$: $s = 0.14$, $y = 0.0196$, $\varepsilon = 0.1804$, $\partial E/\partial w_k = -4\varepsilon s x_k$;

weights $(0.5101; 0.2071; 2.498)$. EX 2023, ReLU, weights $(1; -1; 2)$, same sample: $s = -0.1$, $f' = 0$, **no weight changes**. PZH 2022, threshold, weights $(-0.4; 0.2; 0.5)$, $d = +1$: $s = -0.36$, $y = -1$, $\varepsilon = 2$, weights $(-0.2; 0.34; 0.46)$. EX 2020-3, no bias, $w = (-1; 1)$, $x = (-1; 3)$, $d = -1$, $y = +1$: $\varepsilon = -2$, $w = (-0.8; 0.4)$.

Pitfalls. Factor 2 forgotten or added; sign ($\varepsilon = y - d$ reverses the update); output weight already updated used for δ_j. Lecture 6 convention not checked: use the one of the problem statement, otherwise state the one chosen. Official ZH 2023 solution: writes x_0 instead of x_1 in w_1 (value 0.7 correct). The EX 2023 solution insists: understand the gradient method, the key element is not always f'.

Quiz 06.1, $E = \varepsilon$ or $|\varepsilon|$: with ε, minimizing pushes towards $-\infty$ and the signs cancel out; $|\varepsilon|$ is not differentiable at 0 and its derivative keeps the magnitude ± 1; ε^2 is differentiable, derivative proportional to the error.

3. Decision tree and information gain

Formulas (given in the statement of series 2):

$$I = -\sum_j P(j)\log_2 P(j), \quad I\Big(\tfrac{p}{p+n}, \tfrac{n}{p+n}\Big) = -\tfrac{p}{p+n}\log_2\tfrac{p}{p+n} - \tfrac{n}{p+n}\log_2\tfrac{n}{p+n}$$

$$\text{InfoGain}(A) = I\Big(\tfrac{p}{p+n}, \tfrac{n}{p+n}\Big) - \sum_k \tfrac{p_k+n_k}{p+n} I\Big(\tfrac{p_k}{p_k+n_k}, \tfrac{n_k}{p_k+n_k}\Big)$$

$1/2; 1/2$	1	$1/4; 3/4$	0.811	$0.8; 0.2$ 0.722
$4 \times 1/4$	2	$1/3; 2/3$	0.918	$0.9; 0.1$ 0.469
$1; 0$	0	$2/5; 3/5$	0.971	$1/6; 5/6$ 0.650

n equiprobable outcomes: $\log_2 n$ ($\log_2 3 = 1.585$). $0 \log 0 = 0$. Computing with ln does not change the ranking of the attributes.

Method.

1. Root: count p, n; I_0.
2. Table of branches: examples, $(p_k; n_k)$, I_k (pure: 0; balanced: 1).
3. RemainingInfoNeeded = $\sum_k \frac{p_k+n_k}{p+n} I_k$, **weighted**.
4. Gain = $I_0 -$ remainder; test the attribute with the largest gain (greedy).
5. Impure branch: keep only its examples, start again. Attribute constant on the branch: zero gain.

Results. Series 2 ex. 3 (tennis, 5 Y, 5 N): surface H (3; 0), G (1; 3), C (1; 2), remainder $0.3 \cdot 0 + 0.4 \cdot 0.811 + 0.3 \cdot 0.918 = 0.600$, **gain 0.40 bit**; weather 0.151; height 0.039: root surface, leaf H = Y. ZH 2019 (5000; 5000, five branches): $1 - 0.4 - 0.1 = 0.5$. PZH 2019, four classes: $2 - 0.1 \times 2 = 1.8$. ZH 2020: position, gain $\frac{2}{3}$; branch H: the bishop pair splits, the material advantage (constant) does not. ZH 2022: material advantage, $1 - 0.9387 = 0.061$. ENT 2019: same impure branch $(\frac{1}{4}, \frac{3}{4})$, weight 0.5 against 0.05: gains 0.594 and 0.959.

Pitfalls. Lecture 3 slide 22: "$2 - 4 \times 1 = -2$" is wrong, remainder $= 4 \times \frac{1}{4} \times 1 = 1$, gain 1. Typo in the series 2 statement: $\frac{p}{p+n}$ written twice, the second term is about n. Quiz 03.5 (missed): coin at 80%, $I = 0.722$ bit. n binary attributes: 2^n instances, 2^{2^n} functions (10: 1024 and 2^{1024}); values $k_1, k_2, \ldots : k_1 k_2 \cdots$ instances.

4. Confusion matrix, ROC, AUC

Actual \ predicted	P	N	
P	TP	FN (type II)	sensitivity $TP/(TP+FN)$
N	FP (type I)	TN	specificity $TN/(TN+FP)$
precision $TP/(TP+FP)$; NPV $TN/(TN+FN)$; accuracy $(TP+TN)$/total			

FPR = 1−specificity = $FP/(FP+TN)$. $F_1 = 2PS/(P+S)$, with P the precision and S the sensitivity. Cross-entropy loss: $-\frac{1}{N}\sum_i [y_i \log p(x_i) + (1-y_i)\log(1-p(x_i))]$.

Method (threshold table).

1. Positive class = what is detected (defective, ill); write it down.
2. Totals P and N; "16.917" reads sixteen thousand nine hundred and seventeen.
3. For the threshold: TP and FP = counts on the "positive" side; columns often cumulative.
4. $FN = P - TP$, $TN = N - FP$; check $TP + FN + FP + TN$ = total.
5. Sensitivity, specificity; swapping the positive class swaps them.

Series 2 ex. 1. 23,417 products, 16,917 good, 6,500 defective; above 18: 15,863 good, 697 defective. Positive = defective: TP = 5,803, FN = 697, TN = 15,863, FP = 1,054; sensitivity 0.893, specificity 0.938 (swapped if positive = good). **PZH 2020.** 4 M inhabitants, 5% infected, TPR 0.85, TNR 0.96: 170,000 true flagged, 152,000 false, 322,000 flagged of which 47.2% healthy; precision 0.528. **ENT 2021.** Threshold 30 missing from the table: bracket between columns 25 and 50; threshold where all are positive: TPR 1, TNR 0. **Lecture 3** (20,000 patients, ill if $x \geq 12$): sensitivity 0.8, specificity 0.55, accuracy 0.675, precision 0.64, $F_1 = 0.711$.

ROC: TPR as a function of FPR when the threshold varies. Threshold declaring everything positive: $(1,1)$; nothing: $(0,0)$. Random or irrelevant attribute: diagonal, AUC = 0.5 (quiz 03.6); ideal 1; < 0.5 worse than random. Trapezoids (lecture: 0.754):

$$\text{AUC} \approx \sum_i \frac{(\text{FPR}_{i+1} - \text{FPR}_i)(\text{TPR}_i + \text{TPR}_{i+1})}{2}$$

Typos in the lecture 3 slides. Slide 38: TN in the ill/ill cell (read TP); slide 40: accuracy denominator $TP+FN+FP+FN$ (read $+TN$); slide 39: $(1-y_i)(1-\log(1-p))$ (read $(1-y_i)\log(1-p)$); slide 49: AUC "should be < 0.5" (read > 0.5, as on slide 50). Multiclass: sensitivity of a class on its row, precision on its column. Imbalanced classes: always predicting the majority gives high accuracy and zero sensitivity.

5. K-means

1. Choose K, initialize the centroids.
2. Assignment: table of the d^2 of each point to each centroid (the square root is useless).
3. Update: centroid = mean, coordinate by coordinate, of the assigned points.
4. Repeat until no assignment changes or J iterations. Look for symmetries.

Cost $O(NKJd)$, of which NKd at each iteration. Depends on the initialization; K fixed in advance; clusters assumed spherical and of similar sizes; local optimum.

Series 1 ex. 4 ($c_1 = [5; -1]$, $c_2 = [5; 4]$): cluster 1 $[5;1]$, $[4;-3]$, $[4;0]$, $[7;-1]$; cluster 2 $[5;2]$, $[6;2]$, $[7;7]$, $[2;3]$; $c_1 = [5; -0.75]$, $c_2 = [5; 3.5]$. **EX 2022** ($c_1 = [0;0]$, $c_2 = [6;10]$): only P2 and P9 go to c_2; the other eight are symmetric about the origin: $c_1 = [0;0]$, $c_2 = [6;11]$.

Quiz 04.2 (missed), direction of the move: redo the assignment with the current centroids, then move = cluster mean − old centroid. Do not guess by eye.

Yinyang (triangle inequality): $ub' = ub + d(b,b')$, $lb' = lb - d(c_k, c'_k)$; if $ub' < lb'$, do not compute $d(x, c'_k)$.

6. Activations, vanishing gradient (series 1 ex. 3)

	$f(s)$	$f(0)$	range	f'
sigmoid	$1/(1+e^{-s})$	0.5	$(0;1)$	$y(1-y)$, max 0.25 at 0
tanh	$\frac{e^s - e^{-s}}{e^s + e^{-s}}$	0	$(-1;1)$	$1 - y^2$, max 1 at 0
ReLU	$\max(0, s)$	0	$[0; +\infty)$	0 if $s < 0$, 1 if $s > 0$
threshold	1 if $s \geq 0$, else 0			zero, not differentiable at 0

Sketch: $\sigma(\pm 2) = 0.88$ and 0.12; $\tanh(\pm 1) = \pm 0.76$, $\tanh(\pm 2) = \pm 0.96$; ReLU through $(1;1)$, $(2;2)$, slopes 0 and 1 marked, axes labelled. $\sigma'(0) = 0.25$, $\sigma'(2) = 0.105$, $\sigma'(5) = 0.0066$. Logistic with slope a: $f' = a\,y(1-y)$.

Vanishing gradient, model answer. The gradient of a layer is the product of the derivatives of the following layers. Derivatives < 1: the product tends to 0 with depth, the first layers almost stop learning. Sigmoid and tanh saturate ($f' \approx 0$ for large $|s|$); $\sigma' \leq 0.25$, hence at most $0.25^4 = 0.0039$ over 4 layers, $0.25^5 \approx 0.001$ over 5. ReLU little affected (derivative 1 for $s > 0$), but dead neuron for $s < 0$ (dying ReLU). Lab NN4a: "network paralysis" (sigmoids and very large weights, slide 33); Rprop only uses the sign of the derivative (slide 37). Backpropagation requires a differentiable activation.

7. Hierarchical clustering (series 2 ex. 2)

Bottom-up: N clusters at start, merge of the two closest at each step. Dendrogram: height of a merge = its distance. **Number of clusters at a given distance**: draw the horizontal line, count the vertical lines it crosses. Distance limit: points left alone = anomalies (outliers).

single (minimum)	$\min\{d(a,b) : a \in A, b \in B\}$
complete (maximum)	$\max\{d(a,b) : a \in A, b \in B\}$
average	$\sum_a \sum_b d(a,b) / (\lvert A\rvert \lvert B\rvert)$
centroid	$\lVert c_A - c_B\rVert$

Series 2 ex. 2: **6 clusters** at 2.5 (last merge below the cut near 2.2, next near 2.7); 4 at 3.0; 8 at 2.0 (borderline reading). Unlabelled leaf next to p_8: no effect. Points $0, 1, 3, 7, 12, 13.5$: single gives $\{a,b,c,d\}$, $\{e,f\}$; complete gives $\{a,b,c\}$, $\{d,e,f\}$: the criteria may differ (quiz 04.3, 04.4). **Typo** lecture 4 slide 27: "$d_{S13} < d_{S12}$" for maximum linkage, read $d_{M13} < d_{M12}$. No K to fix; a merge cannot be undone; $O(n^3)$ brute force, $O(n^2 \log n)$ in single linkage. Top-down (divisive): more costly. DBSCAN: ε, MinPts; core point, border point, noise; arbitrary shapes; $O(n^2)$; varying densities handled badly.

8. Distances, K-nearest neighbour, complexity

Euclidean	$\sqrt{\sum_i (x_i - y_i)^2}$; the large magnitude dominates
Manhattan	$\sum_i \lvert x_i - y_i\rvert$; no multiplication
Minkowski	$(\sum_i \lvert x_i - y_i\rvert^p)^{1/p}$; $p = 2$, $p = 1$
cosine	$A \cdot B / (\lVert A\rVert \, \lVert B\rVert)$ (text)
Hamming	positions that differ (binary, categorical)
Mahalanobis	$\sqrt{(\mathbf{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}-\boldsymbol{\mu})}$; diagonal $\boldsymbol{\Sigma}$: $\sqrt{\sum_i (x_i - y_i)^2 / \sigma_i^2}$

Body mass 40 to 150 kg, pH 7.34 to 7.45 (lecture 4 slide 7): Euclidean picks P_2 (4.90 against 5), Mahalanobis P_1 (0.157 against 1.739); uniform distribution, $\sigma = (b-a)/\sqrt{12}$. Quiz 04.1: very different scales, Mahalanobis or scaling.

K-nearest neighbour: K nearest, majority poll; no training, the whole cost at each classification: $O(dN)$ per point. d^2: Nd multiplications, Nd subtractions, $N(d-1)$ additions ($N = 2000$, $d = 10$: 20,000); square root useless; selecting the K: comparisons. On M nodes: best K among MK, total unchanged, time divided by M. K-means $NKdJ$: $2000 \times 7 \times 10 \times 20 = 2.8 \cdot 10^6$. Read quiz 03.1 word for word (multiplications only? square root?).

O: $T(N) = O(F(N))$ if there exist c, N_0 with $T(N) \le c\,F(N)$ for $N > N_0$. $O(1) < O(\log N) < O(N) < O(N \log N) < O(N^2)$. Coins: brute force $N-1$ weighings, halves $\lceil \log_2 N \rceil$, thirds $\lceil \log_3 N \rceil$. Sorting: quick and merge $O(N \log N)$ on average (quick $O(N^2)$ worst case), bubble $O(N^2)$ and memory $O(1)$; quicksort memory: the slide gives $O(N \log N)$, answer with the slide. Dimension: cells 20^{dim}.

9. Generalization, overfitting, data

- Overfitting: model too complex that learns the noise; training error going down, validation error going back up. EX 2020-2 Q6: E5 separates the test set without error, EV makes errors on the test set.
- Validation: stopping, hyperparameters. Test: once, at the end. Choosing on the test set makes the reported error too optimistic.
- Split 60:20:20 (NN2a); 70:15:15 (lecture 5). Each set reproduces the real distribution (20-20-60 example of lecture 3).
- **Early stopping**: stop at the first rise of the validation error. **Backtracking** (recommended): train, save each epoch, go back to the validation minimum. Series 0.40; 0.31; 0.26; 0.24; 0.25; 0.23: epoch 4 against epoch 6.
- More hidden neurons: training error always lower, test error U-shaped. More examples: better generalization.
- Noise on the inputs (jitter): less overfitting; on the targets: only slows down.
- Remedies: small network, much data, early stopping, regularization $\gamma\,\text{mse} + (1-\gamma)\,\text{msw}$.
- Momentum: $\Delta w(n) = \alpha\,\Delta w(n-1) + \eta\,\delta\,y$. Sequential (after each example, preferred) or batch (after each epoch).
- Supervised (labels: K-nearest neighbour, trees); unsupervised (K-means, hierarchical); reinforcement (delayed reward).

10. PCA

1. Centre each column.
2. $\boldsymbol{\Sigma} = \frac{1}{N} \mathbf{X}_{(st)}^\top \mathbf{X}_{(st)}$.
3. Eigenvalues: $\lambda^2 - \operatorname{tr}\lambda + \det = 0$ for 2×2; unit eigenvector.
4. Described variance $\sum_{i \le r} \lambda_i / \sum_i \lambda_i$.
5. Project onto $\mathbf{u}_1, \ldots, \mathbf{u}_r$.

Points $(7,5)$, $(1,-1)$, $(5,1)$, $(3,3)$: mean $(4,2)$, $\Sigma = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, $\lambda = 9$ and 1, $\mathbf{u}_1 = (1,1)/\sqrt{2}$, 90%, projections ± 4.243, 0, 0. SVD: $\lambda_i = s_i^2/N$. Linear only; new variables hard to interpret. Lab 3: `pca_cancer.m` does not centre.

11. Use case: fall detection (lecture 5)

Three-axis accelerometer, 62.5 Hz; normalization $x_{nk} = \frac{x_k - \text{mean}}{\text{std}}$ (std divided by N); overlapping 2 s windows (125 samples per axis); features (magnitude $\sqrt{a_x^2 + a_y^2 + a_z^2}$, G-force $\lvert a\rvert/9.8$, tilt, velocity, standard deviation, skewness, kurtosis, durations); small TinyML network with three classes. 3600 trials, 70:15:15 (2520, 540, 540). Test 561 windows: accuracy float32 $555/561 = 0.989$, int8 $552/561 = 0.984$; no fall missed; int8 divides the weight memory by 4. Anomaly detection: K-means on 9 features, $d(\mathbf{x}) = \min_k(\lVert \mathbf{x} - c_k\rVert_2 - r_k)$, negative or zero: OK; above the threshold: anomaly.

12. Lectures 1 and 2: key facts

Embedded: one task, limited resources (memory, computation, response time, energy), minimal interface, lifetime of ten years or more. EdgeAI: latency, privacy, bandwidth, works without connection. Training is usually not done on the embedded device (ResNet152: 1.5 weeks). Energy (pJ, 45 nm): 32-bit add 0.1; FP32 add 0.9; FP32 multiplication 3.7; 8-bit multiplication 0.2; SRAM 5; DRAM 640 (about 1000 operations). DRAM: $P = \frac{1}{2} \sum C_i V_i^2 f_i$. FP32 $(1, 8, 23)$, FP16 $(1, 5, 10)$, int8 from 0 to 127. **Typo**: the slide gives $6 \cdot 10^4$ for Int16, whereas $2^{15} - 1 = 32{,}767$. Size = parameters × bytes (4, 2, 1): VGG16, 134.2 M, 512 MB in FP32. Brain: 1.3 kg, 20 W, 10^{10} neurons; AlphaGo 1 MW = 50,000 brains. The eight challenges: training time, model size, energy efficiency, inference speed, physical size and weight, safety, trust, governance.